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Blockchain tokenization of tangible assets in Indonesia

Blockchain-Based Tokenization of Tangible Assets in Indonesia: Peer-Reviewed Evidence, Methodologies, and Research Gaps

Quick Reference

Key Findings Table

Asset Class	Tokenization Models & Methods	Legal/Regulatory Status	Technical Challenges	Adoption Factors	Research Gaps
Real Estate/Land	ERC1155C, hybrid blockchain, PKI	Non-recognition of smart contracts, fragmented regulation	Integration with BPN, data integrity, digital literacy	Cognitive factors, management support	Empirical adoption, integration, Sharia compliance
Agriculture/Biological	BEP-20, BEP-721, ERC1155C, parametric insurance	Limited smart contract recognition, regulatory gaps	Stakeholder coordination, scalability, digital literacy	Trust, financial constraints	Legal frameworks, empirical pilots, stakeholder engagement
Natural Resources/Commodities	Asset-backed tokens, REC trading, smart contracts	Regulatory gaps, Sharia compliance, fragmented enforcement	Supply chain integration, data standards, scalability	ESG integration, market readiness	Empirical supply chain studies, harmonization, enforcement
Infrastructure/Energy	P2P trading, REC systems, ERC1155C	Regulatory uncertainty, compliance gaps	Scalability, interoperability, legal enforceability	Stakeholder perceptions, governance	Implementation, standardization, regulatory frameworks
Industrial/Machinery	Debt-/rights-based tokens, supply chain transparency	Legal uncertainty, fragmented regulation	Direct tokenization models, interoperability	Management support, digital literacy	Direct tokenization, empirical pilots, regulatory clarity

Direct Answer

Numerous peer-reviewed studies propose models for blockchain tokenization across real estate, agriculture, natural resources, energy, and industrial assets in Indonesia. However, legal uncertainties, technical integration challenges, and fragmented regulations impede practical implementation. Research has focused on digital ownership, fractional tokenization, and compliance-by-design smart contracts (notably ERC1155C), but empirical data—especially for industrial and machinery asset tokenization—remains scarce. Integration with legacy systems, regulatory harmonization, and stakeholder engagement are urgent research opportunities [1](#) [2](#) [3](#) [4](#) [5](#).

Study Scope

- **Time Period:** 2018–2024
- **Disciplines:** Law, Information Systems, Engineering, Agricultural Economics, Energy Policy, Industrial Management
- **Methods:** Qualitative legal analysis, bibliometric reviews, technical architecture proposals, empirical surveys, case studies, systematic literature reviews

Assumptions & Limitations

- **Assumptions:** Studies focus on tokenization of real-world assets (RWA) and exclude crypto, NFT, DeFi, and traceability-only models.
- **Limitations:** Most research is conceptual or qualitative; empirical and pilot studies are limited, especially for industrial/machinery assets. Regulatory frameworks are evolving and often fragmented [1](#) [6](#).

Suggested Further Research

- Empirical pilot projects for industrial/machinery asset tokenization
- Methodologies for blockchain integration with legacy systems (e.g., BPN, energy grids)
- Regulatory harmonization and Sharia compliance frameworks
- Stakeholder engagement and digital literacy interventions
- Standardized protocols for cross-sector tokenization

1. Introduction

Background and Scope

Blockchain-based tokenization of tangible assets refers to the digital representation of real-world assets (RWA) on distributed ledgers, enabling fractional ownership, asset-backed tokens, and automated transfer of rights via smart contracts. In Indonesia, this approach promises enhanced transparency, efficiency, and democratized access across sectors such as real estate, agriculture, natural resources, infrastructure, and industry [2](#) [3](#) [4](#) [5](#) [7](#). The report

excludes crypto, NFT, DeFi, and traceability-only studies, focusing instead on models that confer ownership, rights, and regulatory compliance.

2. Theoretical Frameworks

Regulatory and Legal Foundations

- **Fragmented Regulation:** Indonesia's legal frameworks for digital assets are fragmented, with non-recognition of smart contracts and lack of integrated digital asset laws [2](#) [8](#).
- **Compliance-by-Design:** Emerging frameworks embed regulatory compliance (KYC/AML, dual approval) directly into smart contracts, notably via ERC1155C extensions [5](#).
- **Sharia Compliance:** Integration of Islamic law principles is increasingly considered, especially for asset-backed tokens and digital gold [9](#) [10](#) [11](#).

Technical Models

- **Hybrid Blockchain Architectures:** Combining permissioned and public blockchains with PKI for secure, scalable asset registries [12](#) [13](#).
- **Smart Contract Automation:** Use of BEP-20, BEP-721, and ERC1155C for fractional ownership and automated asset transfer [3](#) [5](#).
- **Decentralized Storage:** IPFS and similar solutions reduce administrative costs and enhance transparency [5](#).

3. Methods & Data Transparency

- **Legal Analysis:** Juridical-normative reviews of Indonesian statutes, comparative studies with Germany, Malaysia, and Brunei [10](#) [14](#).
- **Technical Proposals:** Blockchain architecture designs, smart contract frameworks, and PKI integration [4](#) [12](#) [15](#).
- **Empirical Surveys:** User adoption studies, stakeholder perception analyses, and readiness assessments [16](#) [17](#).
- **Case Studies:** Pilot implementations in crop insurance, P2P energy trading, and supply chain transparency [3](#) [15](#) [18](#).
- **Systematic Reviews:** Bibliometric and meta-analytical syntheses of blockchain research trends in Indonesia [19](#).

4. Critical Analysis of Findings

4.1 Real Estate and Land Tokenization

Legal and Regulatory Frameworks

- **Non-recognition of Smart Contracts:** Indonesian law partially recognizes digital agreements but lacks authentic deed status, limiting evidentiary strength [20](#).
- **Regulatory Fragmentation:** Multiple agencies oversee digital assets, causing enforcement gaps and legal uncertainty [2](#) [21](#).
- **Need for Harmonization:** Integrated frameworks and reforms are required to support blockchain-based land registries and digital ownership [2](#) [22](#).

Technical Architectures

- **Hybrid Blockchain + PKI:** Proposed for BPN integration, ensuring data authenticity and peer-to-peer security [12](#).
- **Data Integrity Challenges:** Vulnerabilities in centralized systems, legacy data quality, and limited digital literacy impede adoption [12](#) [22](#).

Compliance-by-Design Smart Contracts

- **ERC1155C Framework:** Embeds dual approval (notary, fund manager), whitelist restrictions, and immutable logging for regulatory auditability [5](#).
- **Debt-/Rights-Based Tokenization:** Preferred due to legal restrictions on land title transfer [5](#).

Adoption and User Behavior

- **Cognitive Factors:** Performance expectancy, financial literacy, and perceived risk are key drivers; external factors less significant [16](#) [23](#).
- **Management Support:** Leadership and competitive pressure influence adoption more than technological readiness [23](#).

Research Gaps

- **Empirical Data Scarcity:** Limited quantitative studies on adoption and integration with BPN [12](#).
- **Sharia Compliance:** Insufficient exploration of Islamic law harmonization in digital asset tokenization [24](#).

4.2 Agriculture and Biological Asset Tokenization

Fractional Ownership and Smart Contract Models

- **BEP-20/BEP-721 Tokens:** Used for parametric crop insurance, enabling fractional ownership and automated payouts [3](#).
- **Dynamic Smart Contracts:** Enhance adaptability and scalability in agricultural supply chains [15](#).

Legal and Regulatory Considerations

- **Regulatory Uncertainty:** Smart contracts are used in practice but lack formal legal recognition [2](#) [25](#).
- **Proposed Reforms:** Consolidated digital asset laws and functional asset classification recommended [8](#).

Stakeholder Coordination

- **Fragmented Governance:** Multi-actor involvement, financial constraints, and low blockchain literacy hinder adoption [26](#) [27](#).
- **Trust and Information Asymmetry:** Operational barriers require improved governance and collaboration [28](#).

Case Studies

- **Parametric Crop Insurance:** Blockchain-based models automate payouts, improve transparency, and support cooperative loans [3](#) [29](#).

Research Gaps

- **Legal Frameworks:** Need for smart contract recognition and consumer protection [8](#).
- **Empirical Evaluation:** Scarcity of pilot projects and quantitative adoption studies [30](#).

4.3 Natural Resources and Commodities Tokenization

Legal and Regulatory Frameworks

- **Regulatory Gaps:** Non-recognition of smart contracts, fragmented enforcement, and lack of dispute resolution mechanisms [2](#) [8](#).
- **Sharia Compliance:** Digital gold and ESG assets require integrated legal and Islamic finance frameworks [9](#) [10](#).

Blockchain Applications

- **Commodity Trading:** Asset-backed tokens and REC trading enhance transparency and market efficiency [4](#) [31](#).
- **Intellectual Property:** Blockchain improves IP management, but requires detailed regulations and stakeholder collaboration [32](#) [33](#).

Smart Contracts and Financial Governance

- **Automated Credit Approval:** Smart contracts reduce approval times, increase transparency, and embed compliance rules [34](#) [35](#).

- **ESG Integration:** Blockchain supports sustainable finance and reporting [36](#).

ASEAN-Level Harmonization

- **Regional Coordination:** Proposals for harmonized Sharia-compliant frameworks and cross-border cooperation [10](#) [36](#).

Research Gaps

- **Supply Chain Management:** Need for standardized data protocols and empirical adoption studies [27](#) [37](#).
- **Enforcement and Collaboration:** Weak inter-agency coordination and limited stakeholder engagement [8](#).

4.4 Infrastructure and Energy Asset Tokenization

Peer-to-Peer Energy Trading

- **Empirical Studies:** Positive stakeholder perceptions of blockchain-enabled P2P energy trading platforms in rural Indonesia [17](#).
- **Technical Feasibility:** Platforms like PowerChain demonstrate practical utility and optimization of solar energy [18](#).

Smart Contract Frameworks

- **Compliance-by-Design:** ERC1155C models embed KYC/AML and dual approval for regulatory adherence [5](#) [38](#).
- **Regulatory Sandboxes:** Needed for adaptive legal frameworks and technology-based supervision [39](#).

Carbon Emission Documentation

- **Blockchain-Based Systems:** Enable tamper-proof recording, automated carbon credit trading, and improved compliance [40](#) [41](#).

Scalability and Interoperability

- **Technical Challenges:** Low throughput, transaction delays, and integration difficulties limit efficiency [4](#) [42](#).
- **Legal Enforceability:** Non-recognition of smart contracts and fragmented regulation impede adoption [42](#) [43](#).

Research Gaps

- **Implementation:** Limited practical deployments and standardization efforts [19](#).
- **Regulatory Frameworks:** Need for comprehensive adaptive regulations [43](#).

4.5 Industrial and Machinery Asset Tokenization

Tokenization Models

- **Debt-/Rights-Based Tokens:** Preferred due to legal restrictions; compliance-embedded smart contracts (ERC1155C) proposed [5](#).
- **Supply Chain Transparency:** Blockchain applied mainly for traceability and compliance, not direct asset tokenization [15](#) [44](#).

Regulatory Frameworks

- **Contractual Principles:** Indonesian Civil Code emphasizes mutual consent and substantive justice, impacting smart contract enforceability [45](#).
- **AI-Augmented Compliance:** Governance-aware architectures with human oversight recommended for auditability [46](#).

Adoption Drivers

- **Management Support:** Leadership and competitive pressure are key; digital literacy and regulatory clarity also important [16](#) [23](#).
- **Contextual Factors:** Facilitating conditions and environmental factors influence adoption in MSMEs [47](#) [48](#).

Adapting Compliance Frameworks

- **ERC1155C Adaptation:** Embedding compliance features, regulatory approvals, and immutable records for industrial assets [5](#).
- **Geospatial and AI Integration:** Automates compliance verification and transaction execution [49](#).

Research Gaps

- **Direct Tokenization:** Scarcity of studies on industrial/machinery asset tokenization versus supply chain transparency [6](#) [50](#).
- **Regulatory Clarity:** Need for harmonized laws and pilot projects [51](#).

5. Real-World Implications

- **Legal Certainty:** Tokenization models must align with Indonesian legal doctrines, requiring reforms for smart contract recognition and dispute resolution [2](#) [20](#).
- **Technical Integration:** Hybrid blockchain and PKI architectures are essential for secure, scalable asset registries and integration with legacy systems [12](#) [13](#).

- **Stakeholder Engagement:** Adoption depends on cognitive factors, management support, and digital literacy; targeted interventions are needed [16](#).
- **Regulatory Harmonization:** ASEAN-level coordination and Sharia compliance frameworks can facilitate cross-border tokenization and market trust [10](#) [36](#).
- **Transparency and Efficiency:** Blockchain enhances traceability, reduces fraud, and automates compliance, but requires robust governance and technical solutions [3](#) [31](#) [40](#).

6. Future Research Directions

- **Empirical Pilots:** Launch pilot projects for industrial/machinery asset tokenization and evaluate integration with legacy systems [6](#).
- **Regulatory Reforms:** Develop consolidated digital asset laws, adaptive frameworks, and dispute resolution mechanisms [8](#).
- **Compliance-by-Design:** Expand compliance-embedded smart contract frameworks across sectors, integrating AI and decentralized storage [5](#).
- **Stakeholder Literacy:** Implement training and awareness programs to address digital literacy and trust deficits [27](#) [48](#).
- **Standardization:** Establish cross-sectoral protocols for data integrity, interoperability, and regulatory compliance [27](#) [42](#).

Cross-Sectoral Insights and Synthesis

Common Technical and Regulatory Challenges

- **Scalability and Interoperability:** Integration with legacy systems, data fragmentation, and technical vulnerabilities are recurring issues [8](#) [22](#) [42](#).
- **Legal Uncertainty:** Non-recognition of smart contracts and fragmented regulation impede adoption across asset classes [8](#).

Emerging Solutions and Best Practices

- **Hybrid Blockchain Models:** Combine decentralized ledgers with centralized elements and PKI for authenticity and security [5](#) [35](#).
- **Compliance-by-Design Smart Contracts:** Embed regulatory requirements, dual approval, and KYC/AML into tokenization frameworks [5](#).

- **Stakeholder Engagement:** Leadership, competitive pressure, and targeted training drive adoption [16](#) [23](#).

Opportunities for Regulatory Harmonization

- **ASEAN-Level Coordination:** Harmonized Sharia-compliant frameworks and cross-border cooperation can support tokenization [10](#) [36](#) [39](#).
- **Integrated Legal Frameworks:** Consolidated digital asset laws and adaptive regulations are needed for legal certainty and consumer protection [8](#).

Conclusion

Summary of Findings

Blockchain-based tokenization of tangible assets in Indonesia is conceptually robust, with numerous peer-reviewed studies proposing models for real estate, agriculture, natural resources, energy, and industrial assets. However, practical implementation is hindered by legal uncertainties, technical integration challenges, and fragmented regulatory frameworks. Compliance-by-design smart contract frameworks (notably ERC1155C) and hybrid blockchain architectures are emerging as best practices, but empirical data and pilot projects—especially for industrial/machinery asset tokenization—are lacking. Stakeholder behavior, management support, and digital literacy are critical adoption factors [2](#) [6](#) [12](#) [19](#) [30](#).

Recommendations for Policy and Research

- **Regulatory Reform:** Develop consolidated digital asset laws, recognize smart contracts, and establish dispute resolution mechanisms [5](#) [8](#).
- **Technical Integration:** Invest in hybrid blockchain and PKI architectures for secure, scalable asset registries [5](#).
- **Empirical Pilots:** Launch pilot projects and quantitative studies, especially for industrial/machinery asset tokenization [8](#) [43](#).
- **Stakeholder Engagement:** Implement digital literacy and trust-building programs for users and officials [8](#) [48](#).
- **Cross-Sectoral Standardization:** Harmonize protocols and regulatory frameworks across asset classes and ASEAN jurisdictions [10](#) [39](#).

Synthesis: Indonesia's blockchain tokenization landscape is rich in theoretical models but faces significant practical barriers. Bridging the gap requires regulatory harmonization, technical innovation, and empirical validation, with cross-sectoral adaptation of compliance frameworks and stakeholder engagement as key priorities [1](#) [2](#) [3](#) [4](#) [5](#).

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